

Description and choice data for the domain “Phone plans”

Description of the choice domain 27, Phone plans

The prompt question and the universe of five response options in the choice domain **Phone plans** are as follows. The labels *a*, *b*, *c*, *d* and *e* were not displayed during the experiment and are indicated here to allow cross-referencing with data tables and visualizations below and results in the paper.

The source for this domain is the website of Fido Mobile, a Canadian cell phone carrier, with rates quoted on March 1, 2017 in Canadian dollars.

Of the following cell phone plans, which one would you choose? In all cases, unlimited calling, text picture and video messages to Canadian and international mobile numbers are included. Excess data usage is billed at \$10 per 500 MB.

- 1 GB data per month, \$35 per month.
- 2 GB data per month, \$45 per month.
- 4 GB data per month, \$49 per month.
- 6 GB data per month, \$54 per month.
- 8 GB data per month, \$58 per month.

The following figure is a screenshot from the actual experiment, with one of the 26 possible menus for this domain.

The screenshot shows a web interface titled "Phone plans" in blue. Below the title is a horizontal line. The text "Of the following cell phone plans, which one would you choose? In all cases, unlimited calling, text picture and video messages to Canadian and international mobile numbers are included. Excess data usage is billed at \$10 per 500 MB." is displayed. Below this text are three radio button options: "4 GB data per month, \$49 per month", "1GB data per month, \$35 per month", and "6GB data per month, \$54 per month". To the right of these options is a progress bar with a blue segment on the left. At the bottom right, there is a blue button with the text ">>".

Figure 1: Screenshot for domain Phone plans

Menu A	Choice counts					Choice proportions				
	$N_A(a)$	$N_A(b)$	$N_A(c)$	$N_A(d)$	$N_A(e)$	$\hat{P}_A(a)$	$\hat{P}_A(b)$	$\hat{P}_A(c)$	$\hat{P}_A(d)$	$\hat{P}_A(e)$
$\{a, b\}$	20	20	-	-	-	0.500	0.500	-	-	-
$\{a, c\}$	21	-	19	-	-	0.525	-	0.475	-	-
$\{b, c\}$	-	8	32	-	-	-	0.200	0.800	-	-
$\{a, b, c\}$	12	6	22	-	-	0.300	0.150	0.550	-	-
$\{a, d\}$	19	-	-	21	-	0.475	-	-	0.525	-
$\{b, d\}$	-	20	-	20	-	-	0.500	-	0.500	-
$\{a, b, d\}$	11	8	-	21	-	0.275	0.200	-	0.525	-
$\{c, d\}$	-	-	16	24	-	-	-	0.400	0.600	-
$\{a, c, d\}$	14	-	14	12	-	0.350	-	0.350	0.300	-
$\{b, c, d\}$	-	16	14	10	-	-	0.400	0.350	0.250	-
$\{a, b, c, d\}$	18	4	9	9	-	0.450	0.100	0.225	0.225	-
$\{a, e\}$	20	-	-	-	20	0.500	-	-	-	0.500
$\{b, e\}$	-	15	-	-	25	-	0.375	-	-	0.625
$\{a, b, e\}$	13	9	-	-	18	0.325	0.225	-	-	0.450
$\{c, e\}$	-	-	18	-	22	-	-	0.450	-	0.550
$\{a, c, e\}$	15	-	10	-	16	0.366	-	0.244	-	0.390
$\{b, c, e\}$	-	8	18	-	14	-	0.200	0.450	-	0.350
$\{a, b, c, e\}$	8	10	8	-	14	0.200	0.250	0.200	-	0.350
$\{d, e\}$	-	-	-	23	17	-	-	-	0.575	0.425
$\{a, d, e\}$	20	-	-	9	11	0.500	-	-	0.225	0.275
$\{b, d, e\}$	-	13	-	8	19	-	0.325	-	0.200	0.475
$\{a, b, d, e\}$	19	8	-	7	6	0.475	0.200	-	0.175	0.150
$\{c, d, e\}$	-	-	17	10	13	-	-	0.425	0.250	0.325
$\{a, c, d, e\}$	14	-	11	6	9	0.350	-	0.275	0.150	0.225
$\{b, c, d, e\}$	-	9	19	5	7	-	0.225	0.475	0.125	0.175
$\{a, b, c, d, e\}$	10	8	10	2	11	0.244	0.195	0.244	0.049	0.268

Table 1: Observed choice counts and proportions.

Choice data for domain 27, Phone plans

Table 1 shows choice counts and choice proportions for this choice domain. For each menu A and each object $x \in \{a, b, c, d, e\}$, $N_A(x)$ is the number of participants who chose object x from menu A and $\hat{P}_A(x)$ is the corresponding proportion of participants who chose x from A . When $x \notin A$, a dash is displayed.

The following figure displays choice proportions for all doubleton and tripleton menus in Barycentric coordinates. See a full description of this graphical representation in the paper. Each panel shows choice proportions for all doubleton and tripleton menus of a different tripleton subset of $\{a, b, c, d, e\}$. The downward-pointed (blue) triangle shows the set of ternary choice proportions that are compatible with regularity and the three binary choice proportions, on the corresponding tripleton. The upward-pointed (red) triangle shows the set of ternary choice proportions compatible with the multiplicative inequality and the three binary choice proportions.

Warning in tritrafo(x, y, z): negative components

